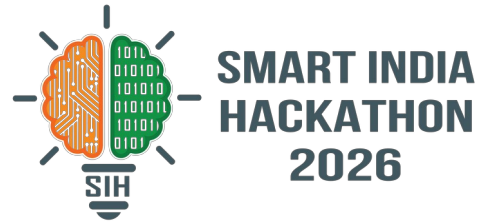


P61	Incredibles	SIH26038	Explaining the Role of AI in the Future of Healthcare	Guruprakash M	6	Graphics and Multimedia Lab (First Floor)- CSE Department
P62	VisionX	SIH26038	Explainable AI for Diabetic Retinopathy Screening Software	Guruprakash M		
P63	VigilanceX	SIH26038	Exploring the Role of AI in the Future of Healthcare	Manisha T		

SMART INDIA HACKATHON 2026



- Problem Statement ID -SIH26038
- Problem Statement Title - Explainable AI for DiabeticRetinopathy Screening in Rural India
- Theme - MedTech / Clean and green technology
- PS Category- Software
- Team ID - P62
- Team Name - visionX



RetinaCare AI – Explainable Diabetic Retinopathy Screening for Rural India

VisionX is an offline, edge-first CDSS that brings explainable DR screening to rural PHCs — without cloud GPUs or continuous internet.

01 5-PILLAR QUALITY GATE

Rejects blur, bad exposure, low coverage and non-retinal images before AI inference.

02 1.2s EDGE INFERENCE

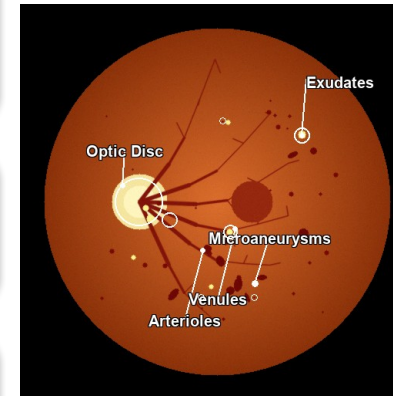
ResNet-50 grades 5 DR severity classes on standard CPU hardware.

03 LESION-LEVEL XAI

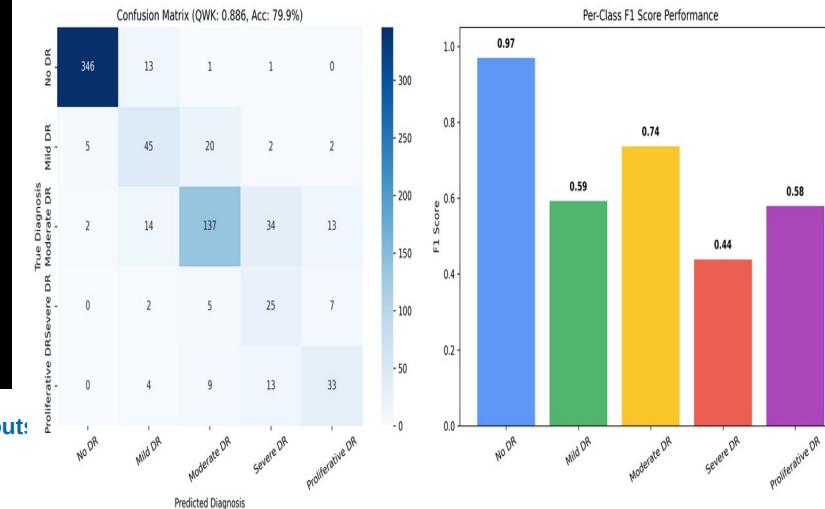
Grad-CAM + optic-disc isolation + lesion localization shows the visual evidence.

04 CLINICAL SAFETY

Calibration preserves severe/PDR predictions to protect against critical misses.

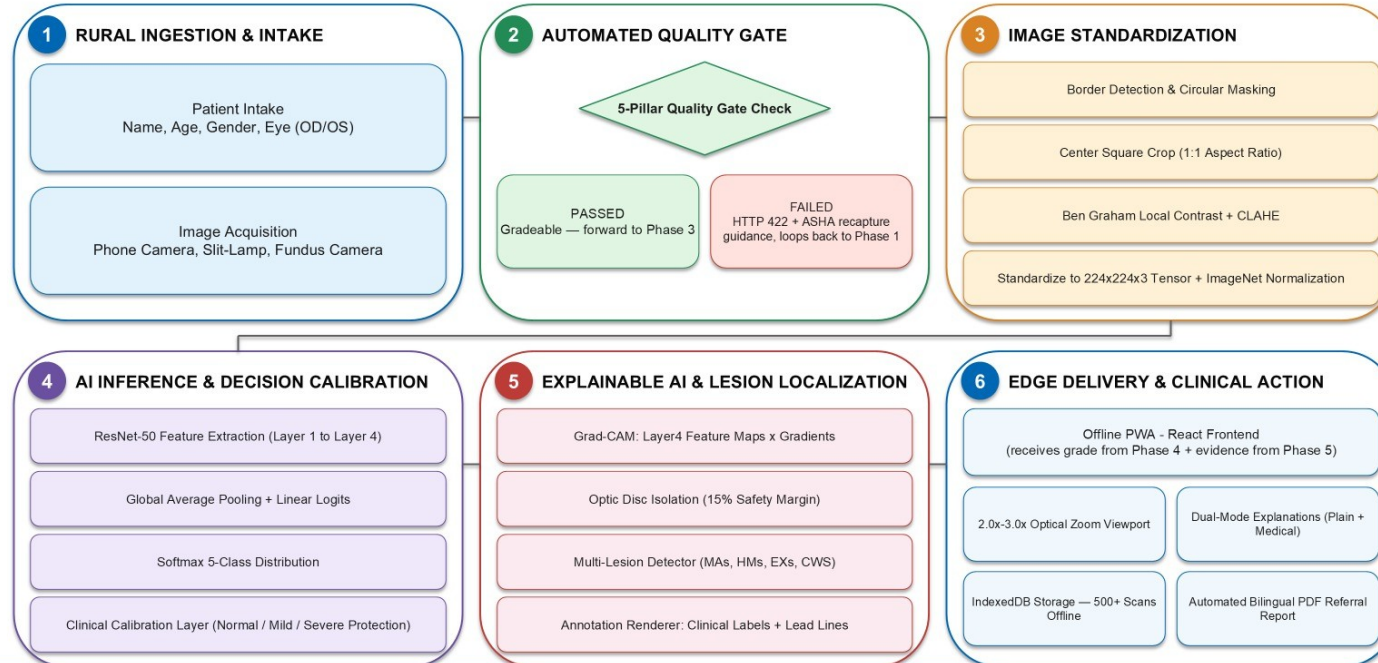


Grad-CAM + lesion callout!



VisionX

DETAILED SYSTEM ARCHITECTURE



VisionX - @SIH Idea Submission

CORE STACK

PyTorch
ModelOpenCV
Quality/XAIFastAPI
Local APIReact
PWAIndexedDB
Offline

MODEL CHOICE

ResNet-50

25.5M params • ~100 MB • 1.2 s CPU

Chosen for rural offline feasibility + native Grad-CAM interpretability.

XAI / LESION ENGINE

Grad-CAM

Layer4 attention map

Optic Disc

15% safety mask

Lesions

MAs • HMs • EXs • CWS

Report

Labels + lead lines + PDF

visionX

FEASIBILITY



Technical Feasibility

QWK 0.8855 • 79.95% exact 5-class accuracy
100% sensitivity on severe + PDR cases.



Operational Feasibility

100% offline PWA + IndexedDB
500+ patient screening records locally.



Economic Feasibility

>99% lower screening triage cost
₹1,500 specialist fee → <₹10 per scan.



Regulatory Feasibility

ICDR severity scale + ETDRS terminology
Designed around clinical grading conventions.

MARKET VIABILITY

77M+ diagnosed diabetics • <25,000 ophthalmologists

Current

77M+

Need

150k+ PHCs

FEASIBILITY AND VIABILITY

CHALLENGES

Rural deployment risks and our fixes

1

Bad input data

Problem: Blurred / poorly-lit / non-retinal photos

Fix: 5-Pillar Quality Gate intercepts bad scans before inference.

2

AI black box

Problem: Doctors cannot verify why a grade was made

Fix: Grad-CAM + lesion-level heatmaps + dual-mode explanations.

3

No internet / GPU

Problem: Rural sub-centers lack bandwidth and cloud hardware

Fix: CPU-optimized PyTorch + browser-cached offline PWA.

MODEL SELECTION

ResNet-50

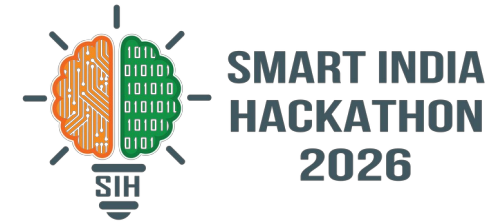
1.2–2.5s • <1.5GB

EfficientNet-B7

10–16s • >4GB

ViT-L/16

25–45s • >8GB



Economic Benefits

Drastically cuts travel and consultation expenses for rural patients.
Empowers low-cost, scalable nationwide screening.

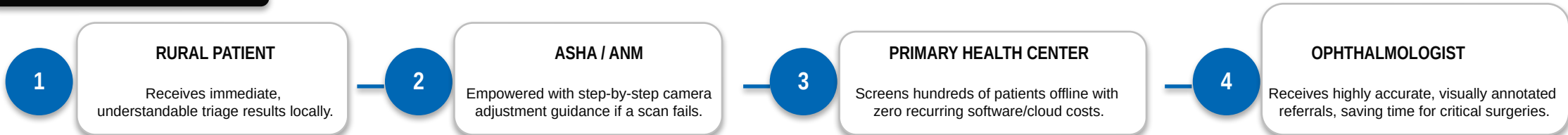
Social Benefits

Democratizes expert ophthalmology access for 77M+ diabetics.
Empowers local ASHA/ANM workers with actionable UI guidance.

Environmental Benefits

Eliminates millions of unnecessary rural-to-urban commutes for healthy patients, significantly reducing the healthcare carbon footprint.

STAKEHOLDERS & IMPACTS



Quality

5-pillar pre-check gate

Blocks blur & bad scans instantly
Live prompts guide ASHA/ANM staff

Trust

Transparent explainable AI

Grad-CAM shows exact pathology
Pinpoints MAs, exudates, bleeds

Reach

100% edge-offline triage

No GPU/internet - 1.2s per scan
Scales to 150k+ rural HWCs

Continuity

Closed-loop care chain

Bilingual PDF report on-spot
Linked to ABHA/district EHR

OUR PROMISE

100% sensitivity on sight-threatening cases (severe and PDR cases).

98% precision on healthy retinas

SDG 3

Good Health & Well-being

SDG 10

Reduced Inequalities

CLINICAL RESEARCH

APTOS 2019

3,662 expert-annotated color fundus images
ICDR Grades 0–4

VALIDATION

QWK 0.8855 • 79.95% exact accuracy
100% severe/PDR sensitivity

INDIA NEED SIGNAL

77M+ diagnosed diabetics
<25,000 ophthalmologists

CLINICAL CONVENTIONS

AAO • ICO • ETDRS
Severity grading terminology

AI / XAI

ResNet-50 — He et al. (2016)

Grad-CAM — Selvaraju et al. (2017)

OUR WORKS

GitHub

Project source + implementation

Project Proof

Validation outputs + generated reports

DATASET / BENCHMARKS

- APTOS 2019 Blindness Detection
- Messidor-2 & IDRI benchmarks

DEEP LEARNING

- He, Zhang, Ren & Sun — Deep Residual Learning
- Selvaraju et al. — Grad-CAM

PREPROCESSING

- Ben Graham — DR competition winner's report
- OpenCV CLAHE / normalization pipeline

CLINICAL GUIDELINES

- American Academy of Ophthalmology (AAO)
- International Council of Ophthalmology (ICO)
- ETDRS Research Group

PROJECT PROOF

Validation split: 733 images • QWK: 0.8855 • Accuracy: 79.95%

github.com/Boobathi-V/SIH_2026.git

Need → Quality → AI → Evidence → Offline Action